TOTAL WEIGHT CHOOSABILITY OF GRAPHS: AN ELEMENTARY NOTE ON COMPLETE MULTI-PARTITE GRAPHS

XIANZHI WANG

Abstract. A graph $G = (V, E)$ is called k-totally weight choosable if the following holds: for any total list assignment L that assigns to each $x \in V \cup E$ a set $L(x)$ of k real numbers, there exists a mapping $f : V \cup E \mapsto \mathbb{R}$ such that $f(x) \in L(x)$ for any $x \in V \cup E$, and for any two adjacent vertices v, u, we have $\sum_{e\in N(v)} f(e) + f(v) \neq \sum_{e\in N(u)} f(e) + f(u)$, where $N(v)$ denotes the edges that incident to v. In this note, we adapt proofs from [2] and [7] to show that the graph join $C_n + \overline{K_m}$, $P_n + \overline{K_m}$, and complete multi-partite graphs are all 3-totally weight choosable.

1. Introduction

All the graphs considered in this note are simple, and have neither isolated edges nor isolated vertices. Let $G = (V, E)$ be a graph. For vertex $v \in V(G)$, let $N(v)$ denotes the set of edges of G that is incident to v. A **weighting** w of G is a function $w : V \cup E \mapsto \mathbb{R}$ that assigns to each edge e (and each vertex v) a real number $w(e)$ (and $w(v)$). Given a weighting w of G, we can define a **total weight** for each vertex $v \in V$ as $t_w(v) := w(v) + \sum_{e\in N(v)} w(e)$. We call the weighting w **neighbor distinguishing** or **proper** if $t_w(u) \neq t_w(v)$ for any edge $\{u, v\}$ in $E(G)$. For a finite set of real numbers $\{1, \ldots, k\}$, we say G is $\{1, \ldots, k\}$-**totally weight colorable** if there is a weighting $w : V \cup E \mapsto \{1, \ldots, k\}$ of G that is neighbor distinguishing. We are mainly interested in the following list version of the problem. Let each edge $e \in E(G)$ and each vertex $v \in V(G)$ be assigned a list (set) of k real numbers, denoted by $L(e)$ and $L(v)$, and let $L = \cup_{x\in V\cup E} L(x)$ denote the collection of all these lists. We refer to such an assignment L as a **total list assignment** L of G and of size k. A graph $G = (V, E)$ is called k-**totally weight choosable** if for any total list assignment L of G and of size k, there is a neighbor distinguishing weighting $f : V \cup E \mapsto \mathbb{R}$ such that $f(x) \in L(x)$ for any $x \in V \cup E$.

The problem of finding the smallest k such that a graph G is k-totally weight choosable is studied quite extensively in the literature, and a survey on this topic is given by Seamone [9]. A slightly different approach is to say a graph $G = (V, E)$ is (k, k')-**total weight choosable** if the following holds: for any list assignment L that assigns to each vertex v a set $L(v)$ of k real numbers, and assigns to each edge e a set $L(e)$ of k' real numbers, there is a neighbor distinguishing weighting $f : V \cup E \mapsto \mathbb{R}$ such that $f(x) \in L(x)$ for any $x \in V \cup E$. This version is studied in [13, 12, 11, 10, 5]. Some related recent developments on this subject could be found in [15, 8, 4, 3].

The **graph join** $G := G_1 + G_2$ of two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ with disjoint vertex sets V_1 and V_2 is the graph union $G_1 \cup G_2$ together with all the edges joining V_1 and V_2. The m-point suspension of C_n (also known as the generalized wheel graph) is defined as the graph join $C_n + \overline{K_m}$, where C_n denotes the cyclic graph with n vertices, and $\overline{K_m}$ denotes the empty graph with m vertices. Also, we let P_n denote a path on n vertices.

In this note, we give a slightly weaker generalization of one result on total weight choosability of wheels by Przybyło and Woźniak in [7]. Namely, we adapt their proof to show that the m-point suspension of C_n, which is the graph join $C_n + \overline{K_m}$, is 3-totally weight choosable. Then the same argument immediately implies that fan graph, (which is $P_n + \overline{K_m}$,) and complete bipartite graphs are also 3-totally weight choosable.

Moreover, we slightly extend the results from [2] and [7] to show that complete multi-partite graphs are 3-totally weight choosable. Although our result on complete multi-partite graphs could be obtained by applying the induction theorem in [2] to a known result in [14] (which says complete multipartite graphs of the form $K_{n,m,1,\ldots,1}$ are $(2, 2)$-total weight choosable), our argument uses algebraic methods involving Combinatorial Nullstellensatz [1] and elementary computation of matrix permanents, while the result in [14] is proven using the "max-min weighting method" introduced in the same paper. In fact, all the results in this note are implied by a general theorem proved in [12], which states that every (nice) graph is $(2, 3)$-choosable.

2. Properties of the permanent of a matrix

Let $M_{k\times\ell}(\mathbb{C})$ denote the set of $k\times\ell$ matrices with each entry being a complex number, and we write $M_k(\mathbb{C})$ in the place of $M_{k\times k}(\mathbb{C})$. Let $per(A)$ denote the permanent of a square matrix A. Recall that the permanent of a matrix $A=(a_{ij})\in M_n(\mathbb{C})$ is defined by

$$per(A)=\sum_{\sigma\in S_n}\prod_{i=1}^{n}a_{i\sigma(i)}$$

We collect some basic results about permanents in the following observation.

Observation 1. Let $A\in M_n(\mathbb{C})$ and $\alpha,\beta\in\mathbb{C}$.

- (1) Permanent is homogeneous of degree n. Let $k\in\mathbb{C}$, we have
$$per(kA)=k^n per(A).$$
- (2) Permanent remains the same for taking the transpose
$$per(A^T)=per(A).$$
- (3) The permanent of a matrix is multi-linear on its row vectors (as well as its column vectors). For $j\in[n]$, if the j-th row C of A is a linear combination of two row vectors $C=\alpha C'+\beta C''$, and A' (respectively, A'') is obtained from A by replacing the j-th row C with C', (respectively, with C''), then
$$per(A)=\alpha per(A')+\beta per(A'').$$
- (4) The permanent is invariant under a permutation of the rows (as well as a permutation of the columns).

Proof. **(1)** By definition, we have

$$\begin{aligned}per(kA)&=\sum_{\sigma\in S_n}\prod_{i=1}^{n}ka_{i\sigma(i)}\\&=\sum_{\sigma\in S_n}k^n\prod_{i=1}^{n}a_{i\sigma(i)}\\&=k^n per(A).\end{aligned}$$

(2) By definition, we have

$$\begin{aligned}per(A^T)&=\sum_{\sigma\in S_n}\prod_{i=1}^{n}a_{\sigma(i)i}\\&=\sum_{\sigma\in S_n}\prod_{i=1}a_{i\sigma^{-1}(i)}\\&=per(A),\end{aligned}$$

where the penultimate step follows because the $a_{i\sigma(i)}$ commutes. We could think of taking the product as a "rearranging trick." The last step follows because as σ runs through S_n, σ^{-1} also runs through S_n.

(3) We prove it for the row case, and the column case follows from (2). Let the j-th row C of A be denoted as $(a_{j1},\dots,a_{jn})$ as usual. Let $C'=(b_{j1},\dots,b_{jn})$, and let $C''=(c_{j1},\dots,c_{jn})$, such that $a_{jk}=\alpha b_{jk}+\beta c_{jk}$ for all $k\in[n]$.

$$\begin{aligned}per(A)&=\sum_{\sigma\in S_n}\left(\left(\prod_{i\neq j}^{n}a_{i\sigma(i)}\right)\cdot\left(\alpha b_{j\sigma(j)}+\beta c_{j\sigma(j)}\right)\right)\\&=\sum_{\sigma\in S_n}\left(\alpha\left(\prod_{i\neq j}a_{i\sigma(i)}\right)\cdot b_{j\sigma(j)}\right)+\sum_{\sigma\in S_n}\left(\beta\left(\prod_{i\neq j}a_{i\sigma(i)}\right)\cdot c_{j\sigma(j)}\right)\\&=\alpha per(A')+\beta per(A'')\end{aligned}$$

(4) This follows from the definition since we have summed over all permutations in S_n.

□

Proposition 2.1. *Let* $A = \begin{bmatrix} X & Z \\ \mathbf{0} & Y \end{bmatrix}$ *be a block upper-triangular matrix in* $M_n(\mathbb{C})$ *with blocks* $X \in M_k(\mathbb{C})$, $Y \in M_\ell(\mathbb{C})$, *where* $k + \ell = n$ *and* $k, \ell > 0$ *are integers.* $Z \in M_{k\times\ell}(\mathbb{C})$, *and* $\mathbf{0}$ *denotes the all zero matrix. Then we have*

$$per(A) = per(X) \cdot per(Y).$$

Proof. The proof uses the fact that the direct product of permutation groups $S_k \times S_\ell$ can be imbedded naturally as a subgroup of $S_{k+\ell}$ via the injective group homomorphism φ:

$$\varphi : S_k \times S_\ell \mapsto T \leq S_{k+\ell} :$$
$$(\tau, \rho) \mapsto \sigma := \begin{cases} \tau \text{ for } i \in \{1, 2, \ldots, k\} \\ \rho \text{ for } j \in \{k+1, k+2, \ldots, k+\ell\} \end{cases},$$

where $T = \varphi(S_k \times S_\ell)$ is the image of φ. For any fixed $\sigma \in S_{k+\ell}$, we claim that terms of the form $\prod_i a_{i\sigma(i)}$ are all zero for $\sigma \notin T$.

We observe that since the entries a_{ij} in matrix A for $i > k$ and $j \leq k$ are all zero, then if a permutation σ sends an element $i \leq k$ to $\sigma(i) > k$, then there must exist an s such that $s > k$ and $\sigma(s) \leq k$, because σ is a bijection on $[k + \ell]$. Hence, this means the term $a_{s\sigma(s)} = 0$, which means $\prod_i a_{i\sigma(i)} = 0$. Hence, for all $\sigma \notin T$, we have $\prod_i a_{i\sigma(i)} = 0$. Thus, we have

$$\begin{aligned} per(A) &= \sum_{\sigma\in S_{k+\ell}} \prod_{i=1}^{n} a_{i\sigma(i)} \\ &= \sum_{\sigma\in T\leq S_{k+\ell}} \prod_{i=1}^{k} a_{i\sigma(i)} \prod_{i=k+1}^{k+\ell} a_{i\sigma(i)} \\ &= \sum_{(\tau,\rho)\in S_k\times S_\ell} \prod_{i=1}^{k} a_{i\tau(i)} \prod_{i=k+1}^{k+\ell} a_{i\rho(i)} \\ &= \sum_{\tau\in S_k} \prod_{i=1}^{k} a_{i\tau(i)} \sum_{\rho\in S_\ell} \prod_{i=k+1}^{k+\ell} a_{i\rho(i)} \\ &= per(X) \cdot per(Y). \end{aligned}$$

where we have abused notation slightly in thinking S_ℓ as permuting the elements $\{k+1, k+2, \ldots, k+\ell\}$. □

Proposition 2.2. *Let* A *be a block upper triangular matrix as follows:*

$$A = \begin{bmatrix} A_1 & * & * & * & \ldots & * \\ \mathbf{0} & A_2 & * & * & & * \\ \mathbf{0} & \mathbf{0} & A_3 & * & & * \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & A_4 & \ldots & * \\ \vdots & & & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \ldots & A_k \end{bmatrix},$$

where each block $A_i \in M_{n_i}(\mathbb{C})$, *and* $\mathbf{0}$ *denotes block matrices with all entries equal to zero, and* $*$ *denotes arbitrary block of matrix. Then we have* $per(A) = \prod_{i=1}^{k} per(A_i)$.

Proof. We use induction on k. The base case $k = 1$ holds automatically. For the inductive step, we apply proposition 2.1 to the following:

$$A' = \left[\begin{array}{cccccc|c} A_1 & * & * & * & \ldots & * & * \\ \mathbf{0} & A_2 & * & * & & * & * \\ \mathbf{0} & \mathbf{0} & A_3 & * & & * & * \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & A_4 & \ldots & * & * \\ \vdots & & & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \ldots & A_\ell & * \\ \hline \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \ldots & \mathbf{0} & A_{\ell+1} \end{array}\right],$$

so we have $per(A') = \left(\prod_{i=1}^{\ell} per A_i\right) \cdot per A_{\ell+1}$, and we are done by induction. □

3. Preliminary set-ups

We need the following theorem due to Noga Alon:

Theorem 3.1 ([1]). *Let $\mathbb{F}$ be a field, and let $f = f(x_1, x_2, \ldots, x_\ell)$ be a polynomial in $\mathbb{F}[x_1, x_2, \ldots, x_\ell]$. Suppose the degree of f equals $\sum_{i=1}^{\ell} k_i$, where each $k_i \in \mathbb{Z}$ is non-negative, and suppose further that the coefficient of $x_1^{k_1} x_2^{k_2} \ldots x_\ell^{k_\ell}$ is non-zero. If $S_1, S_2, \ldots, S_\ell$ are subsets of $\mathbb{F}$ with $|S_i| > k_i$, then there exists $(s_1, s_2, \ldots, s_\ell) \in S_1 \times S_2 \times \ldots \times S_\ell$ such that $f(s_1, s_2, \ldots, s_\ell) \neq 0$.*

A short proof of this theorem could be found here [6].

Let $E(G) = \{e_1, \ldots, e_m\}$ and $V(G) = \{v_1, \ldots, v_n\}$ for $m, n \in \mathbb{Z}^+$. Associate a variable x_e for each $e \in E(G)$ and a variable x_v for each $v \in V(G)$. Let $X_v = x_v + \sum_{e \in N(v)} x_e$. Fix an arbitrary orientation of the edges of G. We define the following polynomial:

$$P(G) = P(x_{v_1}, \ldots, x_{v_n}, x_{e_1}, \ldots, x_{e_m}) = \prod_{\{u,v\} \in E(G)} (X_v - X_u). \tag{1}$$

For different orientations of G, $P(G)$ might differ by a sign, which has no impact for our purposes.

Definition 3.2. Given a polynomial $F \in \mathbb{C}[x_1, \ldots, x_l]$ and any monomial $M = cx_1^{k_1} \ldots x_l^{k_l}$ with $c \neq 0$, let $h(M)$ denote the highest exponent of a variable in M. Let $mind(F)$ denote the **monomial index** and it's defined by

$$mind(F) = \min\{h(M)\},$$

where the minimum is taken over all non vanishing monomials in P.

Now, for a graph $G = (V, E)$, let $l = |E| + |V|$. we define the **total monomial index** of a graph G as $tmind(G) := mind(P(G))$. By Theorem 3.1, we have

Observation 2. If $tmind(G) \leq k$, then G is $(k+1)$-totally weight choosable.

Definition 3.3. Let $K = (k_1, \ldots, k_\ell)$ be a sequence of non-negative integers. Let A be a $m \times \ell$ matrix. Write $A = [A_1, \ldots, A_\ell]$ where $A_1, \ldots, A_\ell$ are columns of A. Let $A(K)$ denote the matrix obtained from A by repeating k_i times each column A_i. Then **the permanent index** of A, denoted by $pind(A)$, is defined as the smallest k for which there exists a sequence $K = (k_1, \ldots, k_\ell)$ of non-negative integers such that the following is satisfied:

- $k_1 + \ldots + k_\ell = m$,
- $perA(K) \neq 0$,
- $k_i \leq k$ for $i = 1, \ldots, \ell$.

Let $\mathbb{C}[x_1, \ldots, x_n]_m$ denote the vector space of homogeneous polynomials of degree m in indeterminate $x_1, \ldots, x_n$ over the field of complex numbers $\mathbb{C}$.

For matrix $A = (a_{ij}) \in M_{m \times n}(\mathbb{C})$, define the polynomial $F_A \in \mathbb{C}[x_1, \ldots, x_n]_m$ by

$$F_A(x_1, \ldots, x_n) := \prod_{i=1}^{m} \sum_{j=1}^{n} a_{ij} x_j.$$

We have the following well-known observation for matrix $A = (a_{ij}) \in M_n(\mathbb{C})$ and $F_A \in \mathbb{C}[x_1, \ldots, x_n]_n$:

Observation 3.

$$per(A) = coe(\prod_{i=1}^{n} x_i, F_A), \tag{2}$$

where the notation on the right-hand-side means the coefficient in front of the monomial term $\prod_{i=1}^{n} x_i$ in expansion of the polynomial F_A.

Proof. This equality can be seen from the following expansion:

$$\prod_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_j =$$

$$(a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n) \times \ldots \times (a_{i1}x_1 + a_{i2}x_2 + \ldots + a_{in}x_n) \times \ldots \times (a_{n1}x_1 + a_{n2}x_2 + \ldots + a_{nn}x_n),$$

because we are selecting one x_i from each multiplicative term above without repetition, so after multiplying out, the coefficients of the individual term $\prod_{i=1}^{n} x_i$ would be $\prod_{i=1}^{n} a_{i\sigma(i)}$, and as we collect terms, we get the sum over all permutations $\sigma \in S_n$. In other words, permanent of matrix A is equal to the coefficient in front of the monomial $x_1 x_2 \ldots x_n$ in the polynomial $F_A(x_1, \ldots, x_n)$. □

Hence, we also have the following lemma as noted in [2], [3], [13], [15].

Lemma 3.4. *For polynomial F_A, the coefficient of $x_1^{k_1}\dots x_n^{k_n}$ is equal to $per(A(K))/(k_1!\dots k_n!)$. Stated concisely, we have*

$$coe(x^K, F_A) = \frac{1}{K!} per(A(K)),$$

where $x^K := x_1^{k_1}x_2^{k_2}\dots x_n^{k_n}$ and $1/K! := 1/(k_1!\dots k_n!)$.

Proof. We are looking for the coefficient in front of the monomial x^K in the polynomial $F_A \in \mathbb{C}[x_1,\dots,x_n]_m$. Since we could have $n < m$, and we would like to express this coefficient using the permanent of $A(K)$, we need the term $1/K!$ since for each $i \in [n]$, we have k_i repetitions of the same indeterminate x_i in the monomial term x^K, so we have "over counted" by $k_i!$, hence the division by $k_i!$ for each i. □

Definition 3.5. Let $G = (V, E)$ be a graph with a fixed orientation, and let $V = \{v_1,\dots,v_n\}, E = \{e_1,\dots,e_m\}, l = n + m$. We define the **associated matrix** of G, denoted $A_G = (a_{ij}) \in M_{m\times l}(\mathbb{C})$, to be the following matrix:

$$a_{ij} = \begin{cases} 1 \text{ if } e_j \text{ is incident with the head of } e_i \text{ or } v_{j-m} \text{ is incident to head of } e_i, \\ -1 \text{ if } e_j \text{ is incident with the tail of } e_i \text{ or } v_{j-m} \text{ is the tail of } e_i, \\ 0 \text{ if } e_j, e_i \text{ are not incident and } v_{j-m} \text{ is not incident to } e_i, \end{cases}$$

where $i \in [m]$ and $j \in [l]$.

We immediately observe the following:

Observation 4.

$$P(G) = F_{A_G}.$$

We have the following lemma from [2] and [7].

Lemma 3.6. *Let G be a graph with a given orientation, and let A_G be its associated matrix. Then*

$$tmind(G) = pind(A_G).$$

Proof. We apply Lemma 3.4, so each monomial term in the expansion of $P(G)$ has $per(A(K))/K!$ as coefficient, and this result follows from Definition 3.3 and Definition 3.2. (To be more specific, $per(A(K))$ nonzero (in the former) corresponds to nonvanishing monomials (in the latter); $k_i < k$ for all i corresponds to taking the maximum over the exponents k_i; and smallest such k corresponds to taking minimum of $h(M)$.) □

We include a lemma that's also proven in [7].

Lemma 3.7 ([7])**.** *Let A be a matrix of of size $m \times \ell$, and let $L \in M_m(\mathbb{C})$ be a square matrix of size m such that each column of L is a linear combination of columns of A. Let n_j denote the number of those columns of L in which the j-th column of A appears with non-zero coefficient. If $per(L) \neq 0$ and $n_j \leq r$ for $j = 1,\dots,\ell$, then $pind(A) \leq r$.*

Proof. Let L_k denote the k-th column of L. By assumption, $L_k = \sum_{j=1}^{\ell} b_{kj}A_j$. We write out the columns of L:

$$L = \left[\sum_{j=1}^{\ell} b_{1j}A_j, \dots, \sum_{j=1}^{\ell} b_{kj}A_j, \dots \sum_{j=1}^{\ell} b_{mj}A_j\right],$$

By Observation 1, we could expand $per(L)$ as follows:

$$per(L) = \sum_{J\in[\ell]^m} \prod_{k=1}^{m} b_{kj_k} per\left([A_{j_1}, A_{j_2}, \dots, A_{j_m}]\right),$$

where $J = (j_1, j_2, \dots, j_m) \in [\ell]^m$. We could think of each index function J as selecting columns from the $m \times \ell$ matrix $A = [A_1, A_2, \dots, A_\ell]$ to obtain the $m \times m$ matrix: $[A_{j_1}, A_{j_2}, \dots, A_{j_m}]$. Now, since by assumption $per(L) \neq 0$, then there exist at least one specific (fixed) index function $J = (j_1,\dots,j_m) \in [\ell]^m$ such that $per\left([A_{j_1}, A_{j_2}, \dots, A_{j_m}]\right) \neq 0$. Observe in matrix $[A_{j_1},\dots,A_{j_m}]$, some columns could be identical. For $j_s \in \{j_1,\dots,j_m\}$, each column A_{j_s} appears at most n_{j_s} times (since n_{j_s} gives the total number of columns of L such that A_{j_s} appears in the linear combination with nonzero coefficients). Since $n_{j_s} \leq r$ for any $j_s \in \{j_1,\dots,j_m\}$, there is at most r identical copies of A_{j_s} in $[A_{j_1}, A_{j_2}, \dots, A_{j_m}]$, hence, by definition of permanent index, we have $pind(A) \leq r$. □

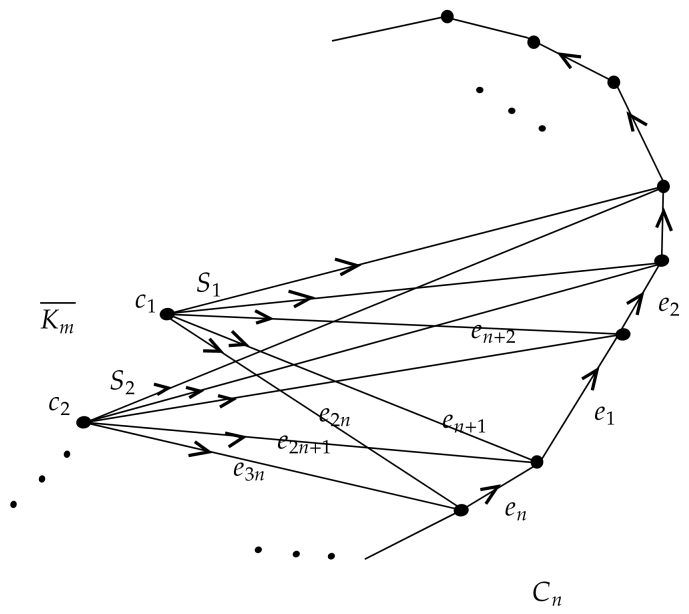

FIGURE 1. Orientation of the edges for m-point suspension of C_n.

4. Application of the Permanent Method

4.1. Application to m-point suspension of C_n, fan graphs, and complete bipartite graphs.

4.1.1. *m-point suspension of C_n.*

Theorem 4.1. *the m-point suspension of C_n, which is the graph join $C_n + \overline{K_m}$, is 3-totally weight choosable.*

We prove this theorem by showing $tmind(G) \leq 2$ where $G = C_n + \overline{K_m}$. We write down the following terminology before starting the proof.

Definition 4.2. By a **block column** we just mean a column in the block matrix, where each entry in the block matrix is a matrix.

Proof. We consider the m-point suspension of C_n, so our cycle has n edges and n vertices, and we have m sets of "spokes", and we call them $S_1, S_2 \ldots, S_m$, along with center points of the "spokes" $c_1, c_2, \ldots, c_m$.

We will follow the notation convention adopted in [7] since the following is a slightly weaker generalization of their proof that the wheel graph is 2-totally weight choosable. We adopt the edge origentation as shown in Figure 1, which the same orientation adopted in [7].

We let A, B, C, D denote the following $n \times n$ matrices, and they will be the "blocks" in our block associated matrices

$$A := \begin{bmatrix} 0 & 1 & 0 & & -1 \\ -1 & 0 & 1 & \ldots & 0 \\ 0 & -1 & 0 & & 0 \\ & \vdots & & \ddots & \vdots \\ 1 & 0 & 0 & \ldots & 0 \end{bmatrix}, B := \begin{bmatrix} -1 & 1 & 0 & & 0 \\ 0 & -1 & 1 & \ldots & 0 \\ 0 & 0 & -1 & & 0 \\ & \vdots & & \ddots & \vdots \\ 1 & 0 & 0 & \ldots & -1 \end{bmatrix},$$

$$C := \begin{bmatrix} 1 & 0 & 0 & & 1 \\ 1 & 1 & 0 & \ldots & 0 \\ 0 & 1 & 1 & & 0 \\ & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \end{bmatrix}, D := \begin{bmatrix} 0 & -1 & -1 & & -1 \\ -1 & 0 & -1 & \ldots & -1 \\ -1 & -1 & 0 & & -1 \\ & \vdots & & \ddots & \vdots \\ -1 & -1 & -1 & \ldots & 0 \end{bmatrix},$$

We let J denote the $n \times n$ matrix with all entries being 1, and we use the bold $\mathbf{0}$ to denote the $n \times n$ matrix with all entries being zero, and we use I to denote the identity matrix as usual.

Also, we let $\vec{0}$ to denote the $n \times 1$ column vector with all 0 entries, and we let $\vec{-1}$ denote the $n \times 1$ column vector with all entries being -1. We also use vtx to indicate the column in the associated matrix that corresponds to the n vertices of the cycle C_n, and we use $cycle$ to indicate (in the associated matrix) the column/row of the n edges correspond to the cycle C_n. Hence, after introducing these notation, we could express the associated matrix $A_{C_n+\overline{K_m}}$ of the graph join $C_n + \overline{K_m}$ as the following block matrix:

$$E := A_{C_n+\overline{K_m}} = \left[\begin{array}{c|cccccccccc} & cycle & vtx & S_1 & S_2 & \dots & S_m & c_1 & c_2 & \dots & c_m \\ \hline cycle & A & B & B & B & \dots & B & \vec{0} & \vec{0} & \dots & \vec{0} \\ S_1 & C & I & D & I & & I & \vec{-1} & \vec{0} & & \vec{0} \\ S_2 & C & I & I & D & \dots & I & \vec{0} & \vec{-1} & \dots & \vec{0} \\ \vdots & \vdots & & & \vdots & \ddots & \vdots & & \vdots & \ddots & \vdots \\ S_m & C & I & I & I & \dots & D & \vec{0} & \vec{0} & \dots & \vec{-1} \end{array}\right]$$

Since we expressed our associated matrix $E := A_{C_n+\overline{K_m}}$ in terms of blocks, each "block column" in this block matrix is in fact n-columns, and again let E_i denote the i-th column of E, we have

$$E = \left[E_1, \dots E_n, E_{n+1}, \dots, E_{2n}, E_{2n+1}, \dots, E_{3n}, E_{3n+1}, \dots, E_{n(m+2)}, \dots, E_{n(m+2)+m}\right].$$

Now we select columns from E and define the following matrix L_E, which is a square matrix, where the number of columns equals to the number of edges in the graph.

$$[E_2, E_3, \dots, E_{n-1}, E_{3n}, E_{2n}, E_{n+1} - E_{2n+1}, \dots, E_{2n-1} - E_{3n-1}, -c_1, S_1 - S_2, S_2 - S_3, S_3 - S_4, \dots, S_{m-1} - S_m]$$

Where E_{2n} is the last column of the block column vtx, and E_{3n} is the last column of the block column S_1. Also, $c_1 = E_{n(m+2)+1}$ denotes the same column. We observe that $-c_1 = E_{2n} - E_{3n}$, and we recognize that $E_{n+1} - E_{2n+1}, \dots, E_{2n-1} - E_{3n-1}$ are exactly the first $n-1$ columns of the block column $vtx - S_1$. Moreover, $S_1 - S_2, S_2 - S_3, S_3 - S_4, \dots, S_{m-1} - S_m$ denotes block columns.

Hence, each column in E is "used" at most twice (in forming linear combination of columns to construct L_E). Since $I - D = J$ where J denotes the all-ones matrix, we could express L_E as the following block matrix:

$$L_E = \left[\begin{array}{c|cccccc} & cycle' & vtx - S_1 & S_1 - S_2 & S_2 - S_3 & \dots & S_{m-1} - S_m \\ \hline cycle & A' & \mathbf{0} & \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} \\ S_1 & C'' & J & -J & \mathbf{0} & \dots & \mathbf{0} \\ S_2 & C' & \mathbf{0} & J & -J & & \mathbf{0} \\ S_3 & C' & \mathbf{0} & \mathbf{0} & J & \dots & \mathbf{0} \\ \vdots & \vdots & & & \vdots & \ddots & \vdots \\ S_m & C' & \mathbf{0} & \mathbf{0} & \mathbf{0} & \dots & J \end{array}\right]$$

Where

$$A' = \left[\begin{array}{cccccc|cc} 1 & 0 & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & \dots & 0 & 0 & 0 \\ \vdots & & & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & \dots & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & \dots & -1 & -1 & -1 \end{array}\right]$$

We do not need to write out the entries of C'' and C' for the purpose of computing the permanent, (although we do know the values) since we will use Laplace expansion along the top rows. A' has 1 on the diagonal, except the last one: $a_{nn} = -1$. Upper triangular part of A' is all zero, except the entry $a_{n-1,n}$ (notice we are selecting columns similar to the techniques in [7]). Therefore, we could compute the permanent explicitly:

$$per(L_E) = 1^{n-2} \cdot (-2) \cdot (n!)^m \neq 0.$$

Note that the columns of L_E are linear combinations of columns of E and none of the columns of E appears more than twice in these linear combinations. Therefore, by lemma 3.7, the permanent index of the associated matrix of $C_n + \overline{K_m}$ is ≤ 2. Thus, by lemma 3.6, we have $tmind(C_n + \overline{K_m}) \leq 2$. □

Thus, the m-point suspension of C_n is 3-totally weight choosable.

For example, when we have 4 sets of "spokes", i.e., when $m = 4$, the associated matrix $A_{C_n+\overline{K_4}}$ is

$$\begin{array}{c|cccccccccc}
 & cycle & vtx & S_1 & S_2 & S_3 & S_4 & c_1 & c_2 & c_3 & c_4 \\
\hline
cycle & A & B & B & B & B & B & \vec{0} & \vec{0} & \vec{0} & \vec{0} \\
S_1 & C & I & D & I & I & I & \vec{-1} & \vec{0} & \vec{0} & \vec{0} \\
S_2 & C & I & I & D & I & I & \vec{0} & \vec{-1} & \vec{0} & \vec{0} \\
S_3 & C & I & I & I & D & I & \vec{0} & \vec{0} & \vec{-1} & \vec{0} \\
S_4 & C & I & I & I & I & D & \vec{0} & \vec{0} & \vec{0} & \vec{-1}
\end{array}$$

And we select columns from the associated matrix to form the following matrix:

$$L_{\overline{C_n}+\overline{K_4}} = \begin{array}{c|ccccc}
 & cycle' & vtx - S_1 & S_1 - S_2 & S_2 - S_3 & S_3 - S_4 \\
\hline
cycle & A' & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\
S_1 & C'' & J & -J & \mathbf{0} & \mathbf{0} \\
S_2 & C' & \mathbf{0} & J & -J & \mathbf{0} \\
S_3 & C' & \mathbf{0} & \mathbf{0} & J & -J \\
S_4 & C' & \mathbf{0} & \mathbf{0} & \mathbf{0} & J
\end{array}$$

4.1.2. *Fan graphs.*

Corollary 4.3. *Fan graph, which is $P_n + \overline{K_m}$, is 3-totally weight choosable.*

We prove this by showing $tmind(G) \leq 2$ when G is a fan graph.

We observe that we could simply delete the n-th column and the n-th row from the associated matrix of the m-point suspension of C_n given previously, then the same column selection and construction of matrix L works, since removing one edge from C_n yields P_n. Hence, $tmind(G) \leq 2$ when $G = P_n + \overline{K_m}$.

4.1.3. *Complete bipartite graphs.*

Corollary 4.4. *Complete bipartite graph is 3-totally weight choosable.*

we prove this by showing $tmind(G) \leq 2$ where $G = K_{n,m}$.

Proof. We observe a complete bipartite graph $K_{n,m}$ is the graph join $\overline{K_n} + \overline{K_m}$, where $\overline{K_n}$ could be thought of as deleting all the edges from the cycle C_n. Hence, using the same orientation as in the case of m-point suspension of C_n, we could readily obtain the associated matrix $A_{\overline{K_n}+\overline{K_m}}$ from before:

$$A_{\overline{K_n}+\overline{K_m}} = \begin{array}{c|ccccccccc}
 & vtx & S_1 & S_2 & \dots & S_m & c_1 & c_2 & \dots & c_m \\
\hline
S_1 & I & D & I & & I & \vec{-1} & \vec{0} & & \vec{0} \\
S_2 & I & I & D & \dots & I & \vec{0} & \vec{-1} & \dots & \vec{0} \\
\vdots & & & \vdots & \ddots & \vdots & & \vdots & \ddots & \vdots \\
S_m & I & I & I & \dots & D & \vec{0} & \vec{0} & \dots & \vec{-1}
\end{array}$$

We observe that $I - D = J$, and we select the columns from the associated matrix of $\overline{K_n} + \overline{K_m}$ in the following way (as indicated above each column) to form matrix L, which is a square matrix with number of columns equal to the number of edges in the graph.

$$L_{\overline{K_n}+\overline{K_m}} = \begin{array}{c|ccccc}
 & vtx - S_1 & S_1 - S_2 & S_2 - S_3 & \dots & S_{m-1} - S_m \\
\hline
S_1 & J & -J & \mathbf{0} & \dots & \mathbf{0} \\
S_2 & \mathbf{0} & J & -J & & \mathbf{0} \\
S_3 & \mathbf{0} & \mathbf{0} & J & \dots & \mathbf{0} \\
\vdots & \vdots & & \vdots & \ddots & \vdots \\
S_m & \mathbf{0} & \mathbf{0} & \mathbf{0} & \dots & J
\end{array}$$

Since this matrix is block diagonal, we have $per\left(L_{\overline{K_n}+\overline{K_m}}\right) = (n!)^m \neq 0$. Now, since each column from the associated matrix $A_{\overline{K_n}+\overline{K_m}}$ is selected to appear at most twice in forming the matrix $L_{\overline{K_n}+\overline{K_m}}$, we have $pind(A_{\overline{K_n}+\overline{K_m}}) \leq 2$ by lemma 3.7, and since $K_{n,m} = \overline{K_n} + \overline{K_m}$, we have $tmind(K_{n,m}) \leq 2$ by lemma 3.6. □

For example, when we have 4 sets of "spokes", i.e., when $m = 4$, the associated matrix $A_{\overline{K_n}+\overline{K_4}}$ is

$$A_{\overline{K_n}+\overline{K_4}} = \begin{array}{c|cccccccccc} & vtx & S_1 & S_2 & S_3 & S_4 & c_1 & c_2 & c_3 & c_4 \\ \hline S_1 & I & D & I & I & I & \vec{-1} & \vec{0} & \vec{0} & \vec{0} \\ S_2 & I & I & D & I & I & \vec{0} & \vec{-1} & \vec{0} & \vec{0} \\ S_3 & I & I & I & D & I & \vec{0} & \vec{0} & \vec{-1} & \vec{0} \\ S_4 & I & I & I & I & D & \vec{0} & \vec{0} & \vec{0} & \vec{-1} \end{array}$$

Moreover, (for $m = 4$,) we obtain the following matrix after selecting columns from the associated matrix above.

$$L_{\overline{K_n}+\overline{K_4}} = \begin{array}{c|cccc} & vtx - S_1 & S_1 - S_2 & S_2 - S_3 & S_3 - S_4 \\ \hline S_1 & J & -J & \mathbf{0} & \mathbf{0} \\ S_2 & \mathbf{0} & J & -J & \mathbf{0} \\ S_3 & \mathbf{0} & \mathbf{0} & J & -J \\ S_4 & \mathbf{0} & \mathbf{0} & \mathbf{0} & J \end{array}$$

Hence, we have $per\left(L_{\overline{K_n}+\overline{K_4}}\right) = (n!)^4 \neq 0$.

4.2. **Application to Complete Multi-partite Graphs.**

Theorem 4.5. *Complete multi-partite graphs are 3-totally weight choosable.*

We need the following (modified) theorem from [2] in order to give a inductive proof that $tmind(G) \leq 2$ when G is a complete multi-partite graph.

Theorem 4.6 ([2])**.** *Let $G = (V, E)$ be a simple graph with $tmind(G) \leq 2$. Let U be a non-empty subset of $V(G)$. Let F be a graph obtained by adding two new vertices u, v to G and joining them to each vertex of U. Let H be a graph obtained from F by joining u and v by an edge. Then $tmind(F) \leq 2$ AND $tmind(H) \leq 2$.*

Proof. The original theorem is in the setting of weight choosability of graphs instead of total weight choosability of graphs, but the original proof works identically for both cases. □

Lemma 4.7. *$tmind(G) \leq 2$ when G is the complete 3-partite graph of the form $K_{n,m,1}$.*

Proof. Let the complete 3-partitle graph be $K_{n,m,1}$, and have partite C with $|C| = n$, partite A with $|A| = m$, and partite $B = \{b\}$. We adopt the edge orientation shown in figure 2, which is similar to the edge orientation used in [7] and [2].

In the following computation for $G := K_{n,m,1}$, without loss of generality and for the purpose of lightening notation, we take $|C| = n = 4$ and $|A| = m = 3$, and $|B| = 1$. We adopt the following idiosyncratic notation. Let b/C be a short hand for row (respectively column) indexing that denote this row (respectively column) is associated with edges that goes from (the only) vertex $b \in B$ to vertices in partite C, and is oriented from b to $c \in C$. Similarly, b/A denotes edges that goes from $b \in B$ to vertices $c \in C$. Again, S_i is the same notation as in section 4.1, where S_i denotes the edges that goes from vertex $c_i \in C$ to partite A, and those edges are oriented from c_i to $a_j \in A$. Also, $|S_i| = m = |A|$. vtx is short for the vertex, and $vtxA$ denotes the vertex set of A, and similarly for $vtxB$ and $vtxC$. The other notations are the same as in section 4.1. We also use the following short hand notation for block matrices:

$$E = [E_1\, E_2\, E_3\, E_4] = \left[\begin{array}{ccc|ccc|ccc|ccc} 1&1&1&0&0&0&0&0&0&0&0&0 \\ 0&0&0&1&1&1&0&0&0&0&0&0 \\ 0&0&0&0&0&0&1&1&1&0&0&0 \\ 0&0&0&0&0&0&0&0&0&1&1&1 \end{array}\right], F = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix} = \left[\begin{array}{cccc} -1&0&0&0 \\ -1&0&0&0 \\ -1&0&0&0 \\ \hline 0&-1&0&0 \\ 0&-1&0&0 \\ 0&-1&0&0 \\ \hline 0&0&-1&0 \\ 0&0&-1&0 \\ 0&0&-1&0 \\ \hline 0&0&0&-1 \\ 0&0&0&-1 \\ 0&0&0&-1 \end{array}\right]$$

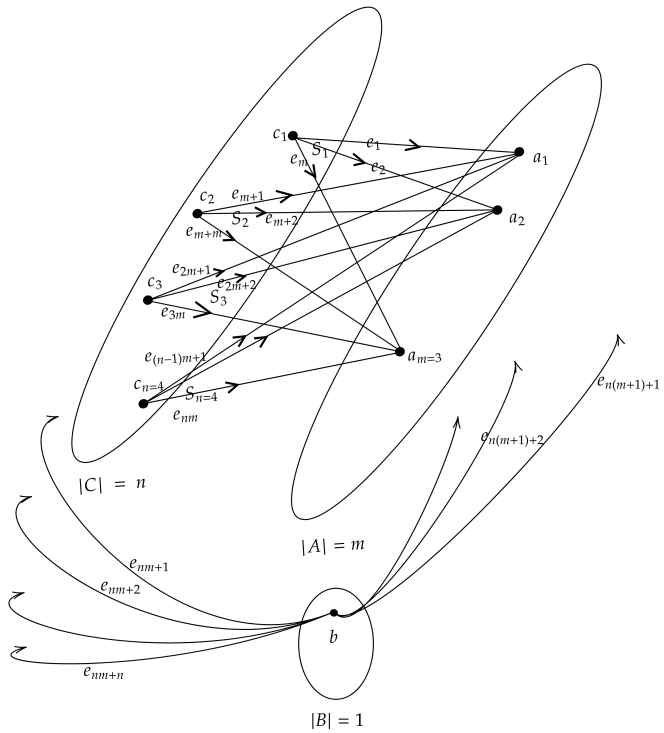

FIGURE 2. Orientation of the edges of $K_{n,m,1}$.

The associated matrix A_G is as follows:

$$A_G = \begin{bmatrix} & vtxA & S_1 & S_2 & S_3 & S_4 & vtxC & b/C & b/A & vtxB \\ S_1 & I & D & I & I & I & F_1 & F_1 & I & \vec{0} \\ S_2 & I & I & D & I & I & F_2 & F_2 & I & \vec{0} \\ S_3 & I & I & I & D & I & F_3 & F_3 & I & \vec{0} \\ S_4 & I & I & I & I & D & F_4 & F_4 & I & \vec{0} \\ b/C & \mathbf{0} & E_1 & E_2 & E_3 & E_4 & I & D & -J & \vec{-1} \\ b/A & I & I & I & I & I & \mathbf{0} & -J & D & \vec{-1} \end{bmatrix}$$

And we select the columns from the associated matrix A_G in the following way,

$$L_G = [vtxA - S_1, S_1 - S_2, S_2 - S_3, S_3 - S_4, vtxC - b/C, vtxA - b/A],$$

and notice we selected each column in A_G at most twice to appear in a linear combination to form L_G. Also, for some block entries, we have put $*$ there instead of writing out the exact value, because the exact value does not play a role in the calculation of the permanent.

$$L_G = \begin{bmatrix} vtxA - S_1 & S_1 - S_2 & S_2 - S_3 & S_3 - S_4 & vtxC - b/C & vtxA - b/A \\ J & -J & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & J & -J & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & J & -J & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & J & \mathbf{0} & \mathbf{0} \\ * & * & * & * & J & J \\ * & * & * & * & J & J \end{bmatrix}$$

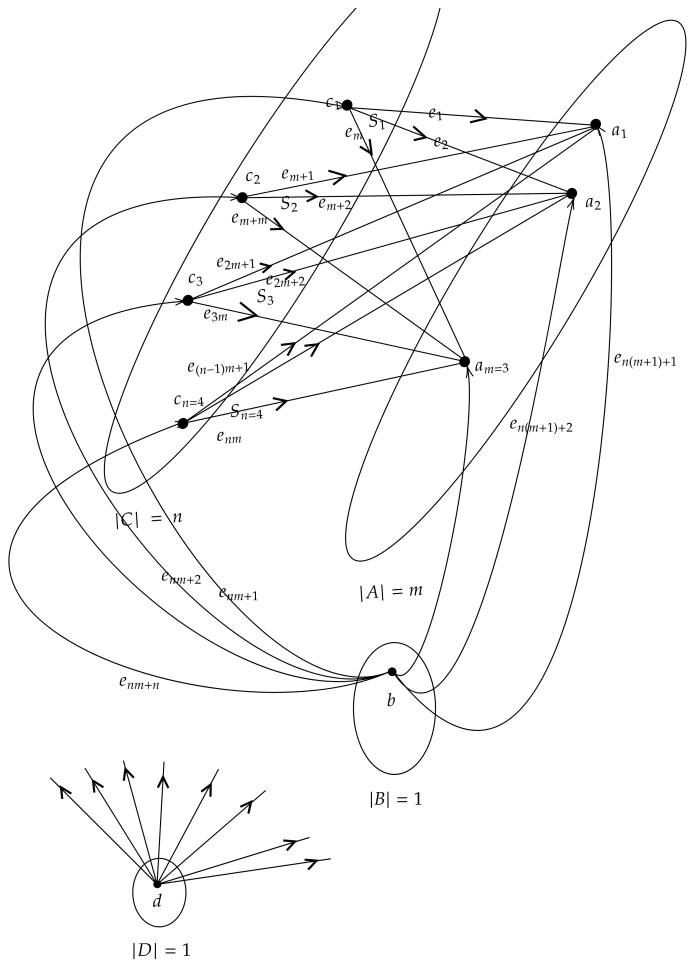

FIGURE 3. Orientation of the edges of $K_{n,m,1,1}$.

Hence, we calculate

$$per(L_G) = per \begin{bmatrix} J & -J & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & J & -J & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & J & -J \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & J \end{bmatrix} \times per \begin{bmatrix} J & J \\ J & J \end{bmatrix} \tag{3}$$

$$= (m!)^n \times (n+m)! \tag{4}$$

$$= (3!)^4 \times (4+3)! \neq 0. \tag{5}$$

Hence, we could apply lemma 3.7 to conclude $pind(A_G) \leq 2$. Hence, by lemma 3.6, we get $tmind(G) \leq 2$. □

Lemma 4.8. *$tmind(G) \leq 2$ when G is the complete 4-partite graph of the form $K_{n,m,1,1}$*

Proof. We use the same set up as in the proof of lemma 4.7 and extend that proof by introducing a new partite consisting of only one vertex, $D = \{d\}$. Hence, we let $G := K_{n,m,1,1}$ with $|C| = n = 4$ and $|A| = m = 3$, $|B| = 1$, and $|D| = 1$ as before. We adopt the edge orientation shown in figure 3, which is similar to the edge orientation used in [7] and [2].

The notations here are the same as in the proof lemma 4.7, and d/b is denoting the single edge going from vertex $d \in D$ to vertex $b \in B$. The associated matrix A_G is as follows:

$$A_G = \begin{bmatrix} & vtxA & S_1 & S_2 & S_3 & S_4 & vtxC & b/C & b/A & vtxB & d/C & d/A & d/b & vtxD \\ S_1 & I & D & I & I & I & F_1 & F_1 & I & \vec{0} & F_1 & I & \vec{0} & \vec{0} \\ S_2 & I & I & D & I & I & F_2 & F_2 & I & \vec{0} & F_2 & I & \vec{0} & \vec{0} \\ S_3 & I & I & I & D & I & F_3 & F_3 & I & \vec{0} & F_3 & I & \vec{0} & \vec{0} \\ S_4 & I & I & I & I & D & F_4 & F_4 & I & \vec{0} & F_4 & I & \vec{0} & \vec{0} \\ b/C & \mathbf{0} & E_1 & E_2 & E_3 & E_4 & I & D & -J & \vec{-1} & I & \mathbf{0} & \vec{-1} & \vec{0} \\ b/A & I & I & I & I & I & \mathbf{0} & -J & D & \vec{-1} & \mathbf{0} & I & \vec{-1} & \vec{0} \\ d/C & \mathbf{0} & E_1 & E_2 & E_3 & E_4 & I & I & \mathbf{0} & \vec{0} & D & -J & \vec{-1} & \vec{-1} \\ d/A & I & I & I & I & I & \mathbf{0} & \mathbf{0} & I & \vec{0} & -J & D & \vec{-1} & \vec{-1} \\ d/b & \vec{0}^T & \vec{0}^T & \vec{0}^T & \vec{0}^T & \vec{0}^T & \vec{0}^T & \vec{1}^T & \vec{1}^T & 1 & \vec{-1}^T & \vec{-1}^T & 0 & -1 \end{bmatrix}$$

We select columns from the associated matrix A_G in the following manner to form matrix L_G:

$$L_G = [vtxA - S_1, S_1 - S_2, S_2 - S_3, S_3 - S_4, vtxC - b/C, vtxA - b/A, b/C - d/C, b/A - d/A, vtxD - d/b],$$

and we have wrote $*$ in the stead of some block entries since the exact value at those entries does not affect calculating the permanent of L_G. Also, the vector arrow $\vec{(\cdot)}$ denotes a column vector of repeated values of $(\cdot)$, and $\vec{(\cdot)}^T$ denotes a transpose of such a vector.

$$L_G = \begin{bmatrix} vtxA - S_1 & S_1 - S_2 & S_2 - S_3 & S_3 - S_4 & vtxC - b/C & vtxA - b/A & b/C - d/C & b/A - d/A & vtxD - d/b \\ J & -J & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \vec{0} \\ \mathbf{0} & J & -J & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \vec{0} \\ \mathbf{0} & \mathbf{0} & J & -J & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \vec{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & J & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \vec{0} \\ * & * & * & * & J & J & -J & -J & \vec{1} \\ * & * & * & * & J & J & -J & -J & \vec{1} \\ * & * & * & * & \mathbf{0} & \mathbf{0} & J & J & \vec{0} \\ * & * & * & * & \mathbf{0} & \mathbf{0} & J & J & \vec{0} \\ * & * & * & * & \vec{-1}^T & \vec{-1}^T & \vec{2}^T & \vec{2}^T & -1 \end{bmatrix}$$

We could see that the above matrix is in 2×2 block diagonal form, where the 2 square matrices on the diagonal are

$$\text{block}_1 = \begin{bmatrix} J & -J & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & J & -J & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & J & -J \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & J \end{bmatrix}, \text{block}_2 = \begin{bmatrix} J & J & -J & -J & \vec{1} \\ J & J & -J & -J & \vec{1} \\ \mathbf{0} & \mathbf{0} & J & J & \vec{0} \\ \mathbf{0} & \mathbf{0} & J & J & \vec{0} \\ \vec{-1}^T & \vec{-1}^T & \vec{2}^T & \vec{2}^T & -1 \end{bmatrix}$$

We know from proof of lemma 4.7 that $per(\text{block}_1) \neq 0$, so we only need to show $per(\text{block}_2) \neq 0$ as well.

Claim: $per(\text{block}_2) \neq 0$.

Since permanent is invariant under permutations of rows and columns, we first cyclically permute the rows so that the last row becomes the first row, then we cyclically permute the columns, so the last column becomes the first column, and we have

$$per(\text{block}_2) = per \begin{bmatrix} -1 & \vec{-1}^T & \vec{-1}^T & \vec{2}^T & \vec{2}^T \\ \vec{1} & J & J & -J & -J \\ \vec{1} & J & J & -J & -J \\ \vec{0} & \mathbf{0} & \mathbf{0} & J & J \\ \vec{0} & \mathbf{0} & \mathbf{0} & J & J \end{bmatrix}$$

$$= (-1) \cdot per \begin{bmatrix} J & J & -J & -J \\ J & J & -J & -J \\ \mathbf{0} & \mathbf{0} & J & J \\ \mathbf{0} & \mathbf{0} & J & J \end{bmatrix} + (-1)(m+n) \cdot per \begin{bmatrix} J & J & -J & -J \\ J & J & -J & -J \\ \mathbf{0} & \mathbf{0} & J & J \\ \mathbf{0} & \mathbf{0} & J & J \end{bmatrix} + 2(m+n) \cdot per \begin{bmatrix} J & J & -J' & -J \\ J & J & -J' & -J \\ \mathbf{0} & \mathbf{0} & J' & J \\ \mathbf{0} & \mathbf{0} & J' & J \end{bmatrix}$$

where we have used Laplace expansion on the first row. In the last matrix, the block column

$$\begin{bmatrix} -J' \\ -J' \\ J' \\ J' \end{bmatrix} \text{ is obtained by replacing the first column of } \begin{bmatrix} -J \\ -J \\ J \\ J \end{bmatrix} \text{ with } \begin{bmatrix} \vec{1} \\ \vec{1} \\ \vec{0} \\ \vec{0} \end{bmatrix}. \tag{6}$$

Hence, we have

$$per \begin{bmatrix} J & J & -J' & -J \\ J & J & -J' & -J \\ \mathbf{0} & \mathbf{0} & J' & J \\ \mathbf{0} & \mathbf{0} & J' & J \end{bmatrix} = per \begin{bmatrix} J & J \\ J & J \end{bmatrix} \times per \begin{bmatrix} J' & J \\ J' & J \end{bmatrix} = (m+n)! \times 0,$$

since the block $\begin{bmatrix} J' & J \\ J' & J \end{bmatrix}$ has a column of zeros. Also, we have

$$per \begin{bmatrix} J & J & -J & -J \\ J & J & -J & -J \\ \mathbf{0} & \mathbf{0} & J & J \\ \mathbf{0} & \mathbf{0} & J & J \end{bmatrix} = per \begin{bmatrix} J & J \\ J & J \end{bmatrix} \times per \begin{bmatrix} J & J \\ J & J \end{bmatrix} = ((m+n)!)^2 \neq 0,$$

where we notice that the block $\begin{bmatrix} J & J \\ J & J \end{bmatrix}$ is of dimension $(m+n) \times (m+n)$, where we have abused notation slightly by using J to denote block matrices of all ones of different dimension. Hence, this together establishes the claim that $per(\text{block}_2) \neq 0$.

Thus, $per(L_G) = per(\text{block}_1) \times per(\text{block}_2) \neq 0$. By lemma 3.7 and lemma 3.6, we conclude $tmind(G) \leq 2$ where $G = K_{n,m,1,1}$.

□

Corollary 4.9. *$tmind(G) \leq 2$ when G is the complete $(k+2)$-partite graph of the form $K_{n,m,1,\ldots,1}$*

Proof. By lemma 4.7 and lemma 4.8, we know that $K_{n,m,1}$ and $K_{n,m,1,1}$ have total monomial index at most 2, we could apply Theorem 4.6 to immediately obtain that complete $(k+2)$-partite graph of the form $K_{n,m,1,\ldots,1}$ also have total monomial index at most 2, by introducing 2 vertices at a time in the induction step. □

Corollary 4.10. *$tmind(G) \leq 2$ when G a complete multi-partite graph.*

Proof. By Corollary 4.9 and Corollary 4.4, the base cases $K_{n,m,1,\ldots,1}$ and $K_{n,m,0,\ldots,0}$ have total monomial index at most 2, so we could apply theorem 4.6 to immediately obtain that complete multi-partite graphs have total monomial index at most 2. □

Hence, Theorem 4.5 follows.

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Middlebury College, Middlebury, VT 05753, United States
Email address: xianzhiw@middlebury.edu